Computer Vision
Lecture 7: Camera Sensor

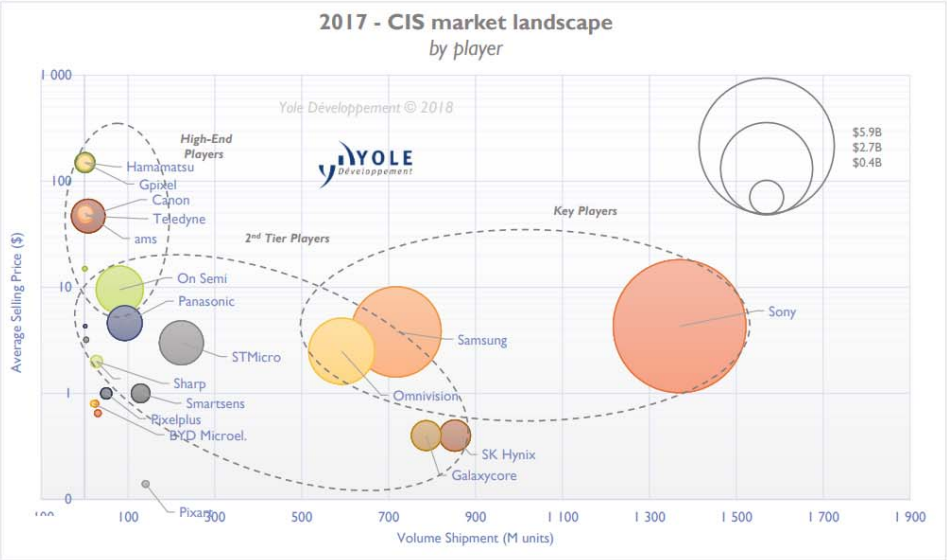
Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

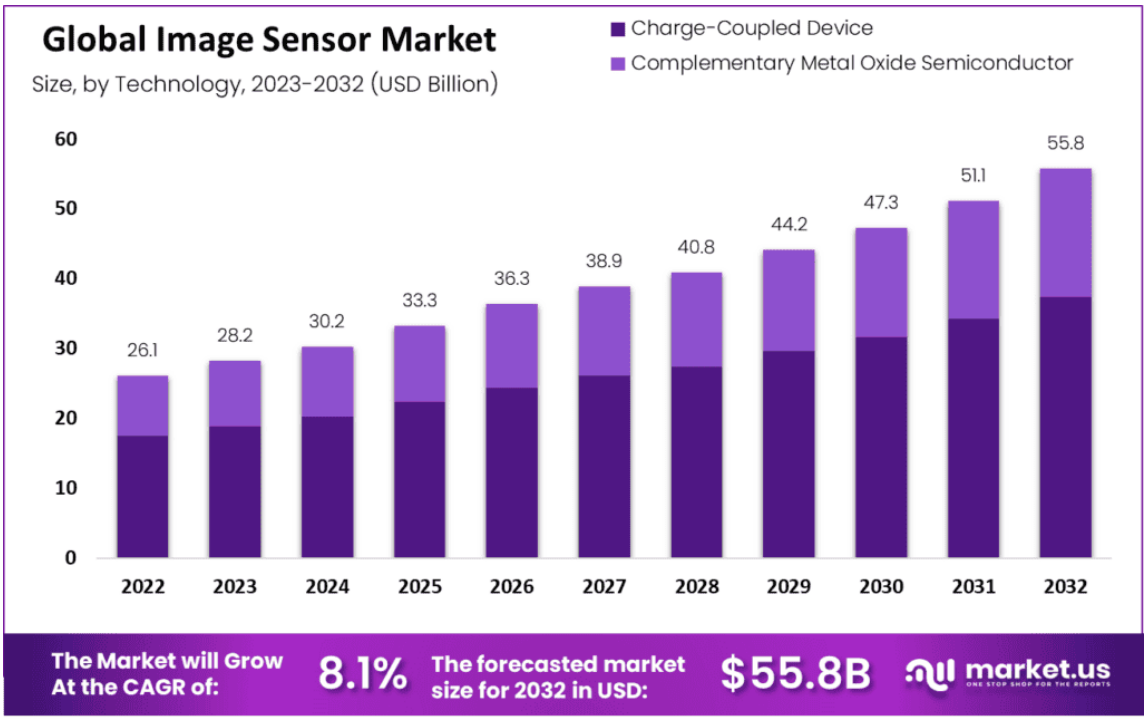
Camera Sensor Market by Player

Sony maintains its market leadership, leveraging technology rooted in digital photography and applied to mobile

Second tier players become more homogenous as On Semi, Panasonic and STMicro maintain growth in respect to Omnivision and Samsung

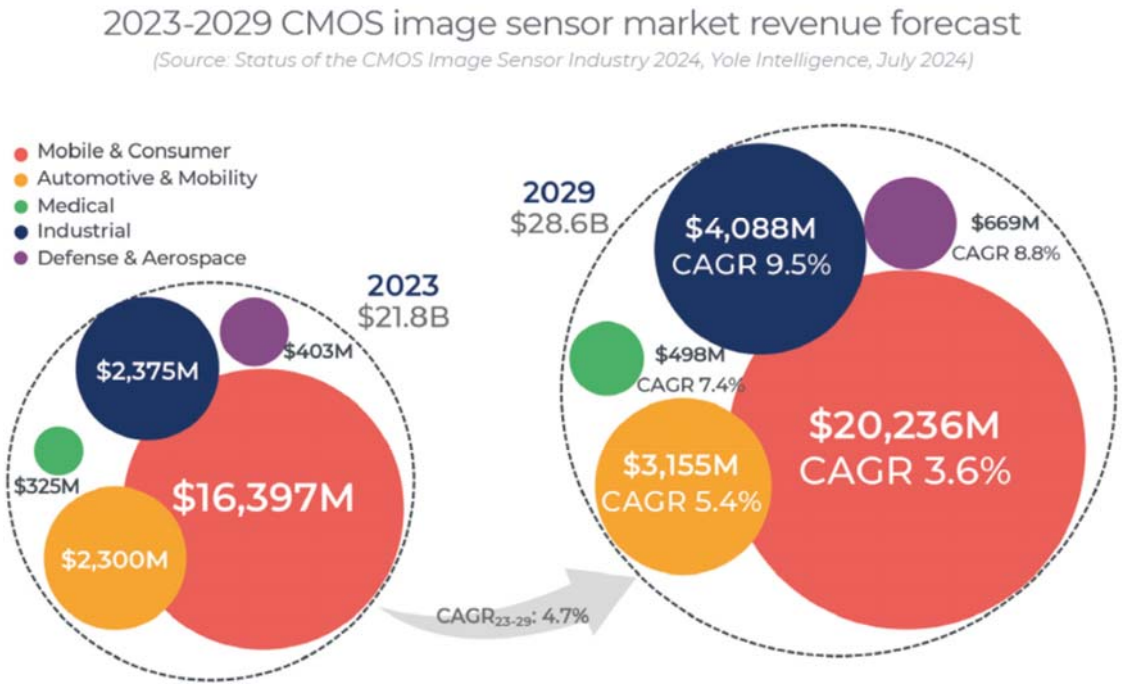


Global Image Sensor Market



CMOS Sensor Market by Industry

- 2017: Mobile > Computing > Consumer > Security > Automotive
- 2023: Mobile > industrial>Automotive > …..



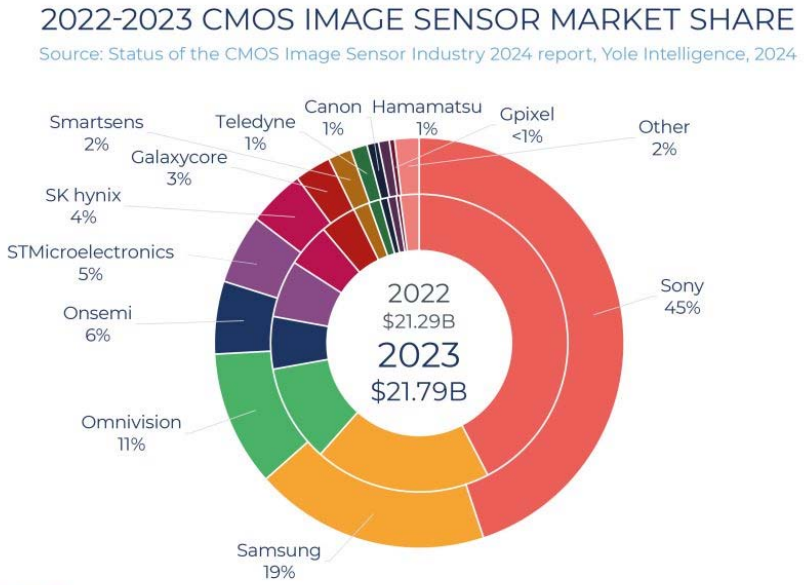
CMOS Sensor Market by Player

- 2023: Sony > Samsung > Omnivision > On Semiconductor > STM ..

Note: On Semi merged Aptina in 2014

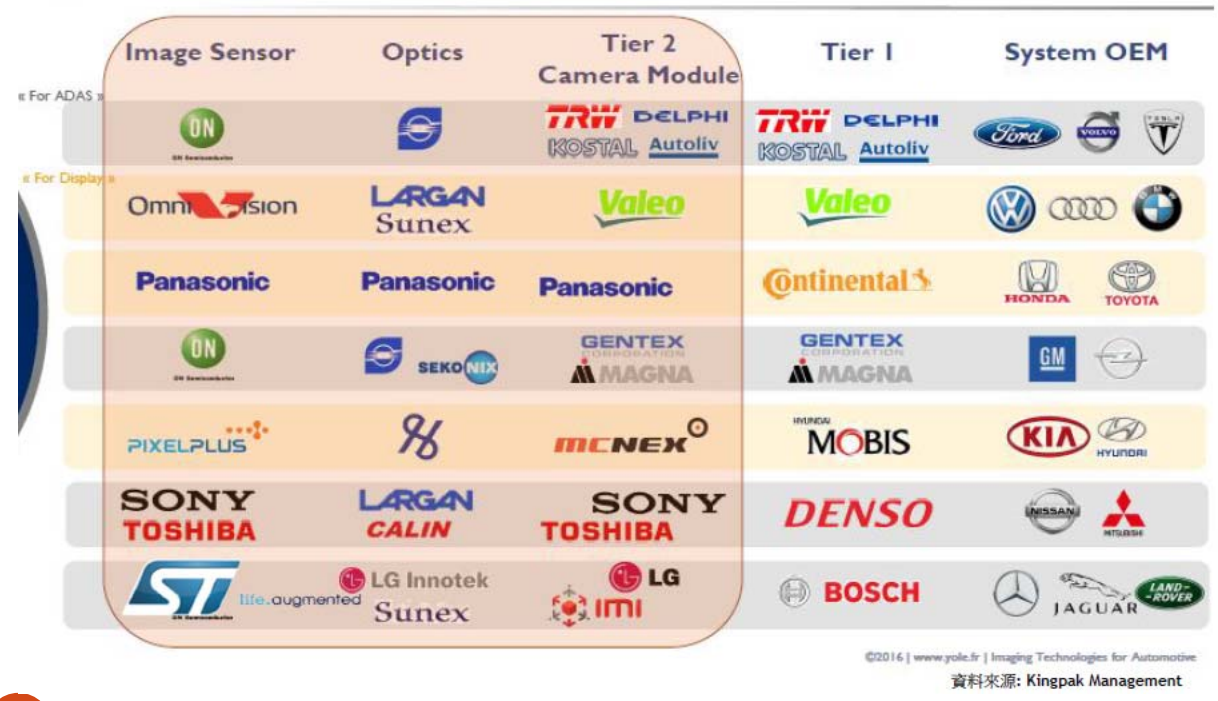
Actual Automotive CMOS Company: Sony, Omnivision, On Semi

https://semiconductor.samsung.com/image-sensor/automotive-image-sensor/



Con

Automotive CMOS Supply Chains

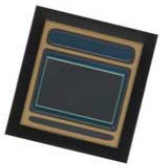


Computer Vision

Example: Sony IMX390 CMOS senor

IMX390CQV

1/2.7-type **2.45megapixels CMOS image sensor**



1. Industry First* simultaneous LFM and HDR
   Realizes simultaneous LFM and HDR functions by using Sony's original pixel architecture and exposure methods. LED traffic signs and LED lights of vehicles can be accurately recognized while enabling HDR image capture with no highlight blowout or loss of shadow detail in situations of dramatic contrast, such as when entering or exiting tunnels.
2. High Sensitivity
   Achieves highest sensitivity in 2.0 megapixel class automotive cameras* to capture high-quality color images even in low light conditions such as 0.1 lux, which is equivalent to moonlight.
3. Meeting quality standards for automotive camera
   - AEC-Q100 Grade 2
   - Development process complies with ISO26262
   - ASIL C functional safety requirements

Product Features

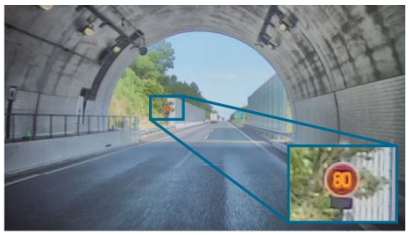
- Number of recommended recording pixels:
  WUXGA 1920 (H) × 1200 (V) approx. 2.30M pixel
  Full-HD 1920 (H) × 1080 (V) approx. 2.07M pixel
- Readout rate
  Maximum frame rate in Full HD 1080P mode:
  60 fps@AD10bit, 40fps@AD12bit
- High dynamic range (HDR) function
- LED Flicker mitigation (LFM) function
- Variable-speed shutter function
  (resolution 1H units)
- 10-bit / 12-bit A/D converter on chip
- CDS / PGA function
- Interface
  MIPI CSI-2 serial output (2lane/4lane)
- AEC-Q100 Grade 2
- ISO26262 / ASIL C

Example: Sony IMX390 CMOS senor

60 fps@AD10bit, 40fps@AD12bit

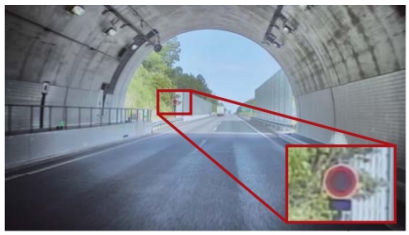
- High dynamic range (HDR) function
- LED Flicker mitigation (LFM) function

Image captured with "IMX390"



Simultaneous LFM and HDR

Image captured with "IMX290"



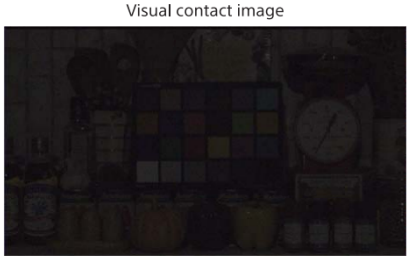
HDR function only

Image captured with "IMX390"



IMX390 under 0.1 lux light condition

Visual contact image



Visual image under 0.1 lux light condition

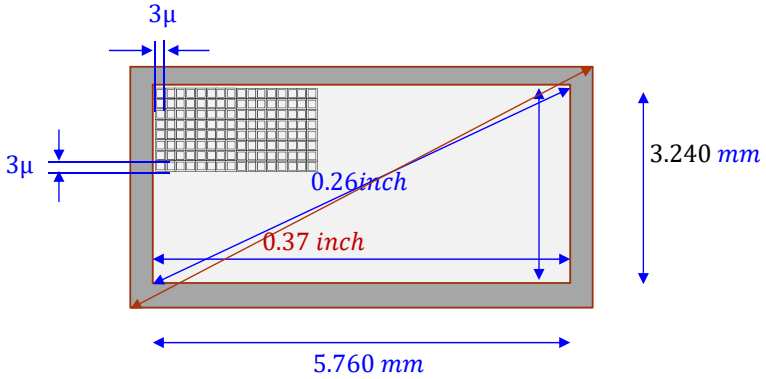
Example: Sony IMX390 CMOS senor

Model name		IMX390CQV
Number of effective pixels		2017 (H) x 1217 (V) 2.45 megapixels
Image size		Diagonal 6.67mm (type 1/2.7)
Unit cell size		3.0μm (H) x 3.0μm (V)
Frame rate	WUXGA	10bit 50fps, 12bit 30fps
	Full HD	10bit 60fps, 12bit 40fps
Dynamic range (EMVA1288 standard)		120dB
Sensitivity (F5.6 standard value, 1/30 second storage time)		1953mV (green pixel)
Saturation signal (minimum value)		700mV
Power supply	Analog	2.9V
	Digital	1.2V
	Interface	1.8V
Interface		MIPI CSI-2 serial output (2lane/4lane)
Package		96pin plastic BGA
Package size		10.0mm x 10.6mm

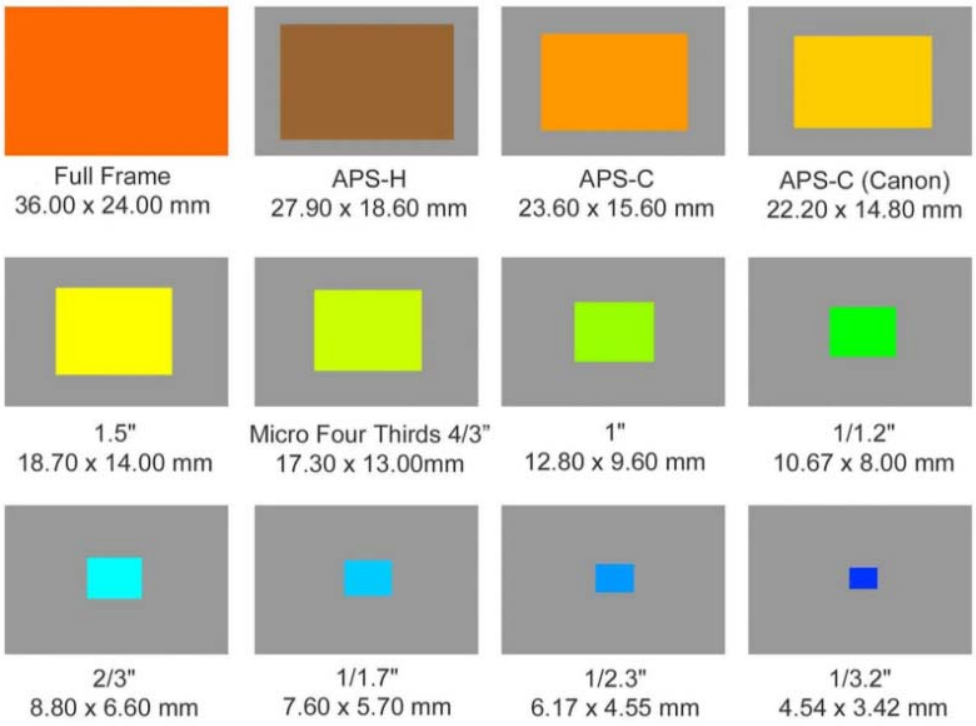
Sensor size

- FHD로 가정 (1,920 x 1,080), 1inch = 25.4 mm
- 가로: $3\mu \times 1{,}920 = 5.760mm$
- 세로: $3\mu \times 1{,}080 = 3.240mm$
- 대각선 : $6.608mm = 0.26\ inch$ ⬅➡ Type $1/2.7" = 0.37\ inch$

Number of effective pixels	2017 (H) x 1217 (V) 2.45 megapixels
Image size	Diagonal 6.67mm (type 1/2.7)
Unit cell size	3.0μm (H) x 3.0μm (V)



Sensor size



Sensor size chart

https://www.studiobinder.com/blog/camera-sensor-size/

Example : On Semi CMOS Sensor

AR0231ATSC00XUEA0-DRBR-E

- Type : CMOS
- Megapixel : FHD 1920 x 1080 = 2,073,600 (약 2 Megapixel)
- Frame Rate (fps) : 30
- Shutter Type : Rolling Shutter
- HDR, Automotive

Product	Compliance	Status	Description	Type	Megapixels	Frame Rate (fps)	Optical Format	Shutter Type	Pixel Size (µm)	Output Interface	Color	Package Type
AR0230ATSC00XUEA0-DRBR	AEC Qualified PPAP Capable Pb-free Halide free	Active	2 MP 1/3" CMOS Image Sensor									IBGA-80
AR0230ATSC00XUEA0-DRBR-E	Pb-free Halide free	Active	2 MP 1/3" CMOS Image Sensor	CMOS	2.1	30	1/2.7 inch	Electronic Rolling and Global Reset Release	3.0 x 3.0	Parallel HiSPi™	RGB	IBGA-80
AR0230ATSM00XUEA0-DRBR-E	Pb-free Halide free	Active	2 MP 1/3" CMOS Image Sensor	CMOS	2.1	30	1/2.7 inch	Electronic Rolling and Global Reset Release	3.0 x 3.0	HiSPi™ Parallel	Mono	IBGA-80

Example : Ambarella ISP+CV: CV22AQ

Overview

Ambarella's AEC-Q100 qualified CV22AQ SoC combines image processing, 8MP30 video encoding, and CVflow® computer vision processing in a single, low-power design. The CV22AQ's CVflow architecture provides the deep neural network (DNN) processing required by the next generation of intelligent automotive cameras. Fabricated in advanced 10 nm process technology, it achieves an industry-leading combination of low-power and high-performance in both human vision and computer vision applications. It is an ideal platform for implementing ADAS, electronic mirror, surround view, and drive recorder solutions.

The CV22AQ's CVflow architecture provides computer vision processing at 8MP resolution, enabling image recognition over long distances and with high accuracy. It includes efficient encoding in both AVC and HEVC video formats, delivering high-resolution video streaming with very low bit rates.
The CV22AQ's next-generation image signal processor (ISP) provides outstanding imaging in low-light conditions while high dynamic range (HDR) processing extracts maximum image detail in high-contrast scenes, further enhancing the computer vision capabilities of the chip. It includes a suite of advanced security features to prevent hacking, such as secure boot, TrustZone® and I/O virtualization. A complete set of tools is provided to help customers easily port their own neural networks onto the CV22AQ SoC.



The CV22AQ chip targets automotive sensing camera designs

Key Features

Computer Vision Engine CVflow®
- CNN- / DNN-based processing: detection, classification, tracking, and more
- Computer vision processor
- Tools for high- and low-level algorithm development
- CNN toolkit for easy porting with Caffe, TensorFlow, and ONNX
- Open SDK

Advanced Image Processing
- Multi-exposure line-interleaved HDR
- Hardware dewarping engine support
- Electronic image stabilization (EIS)
- Support for multiple cameras
- LED flicker mitigation
- 3D motion-compensated temporal filtering (MCTF)
- Superior low-light processing
- Support for RGGB, RCCB, RCCC, RGB-IR, and monochrome sensors

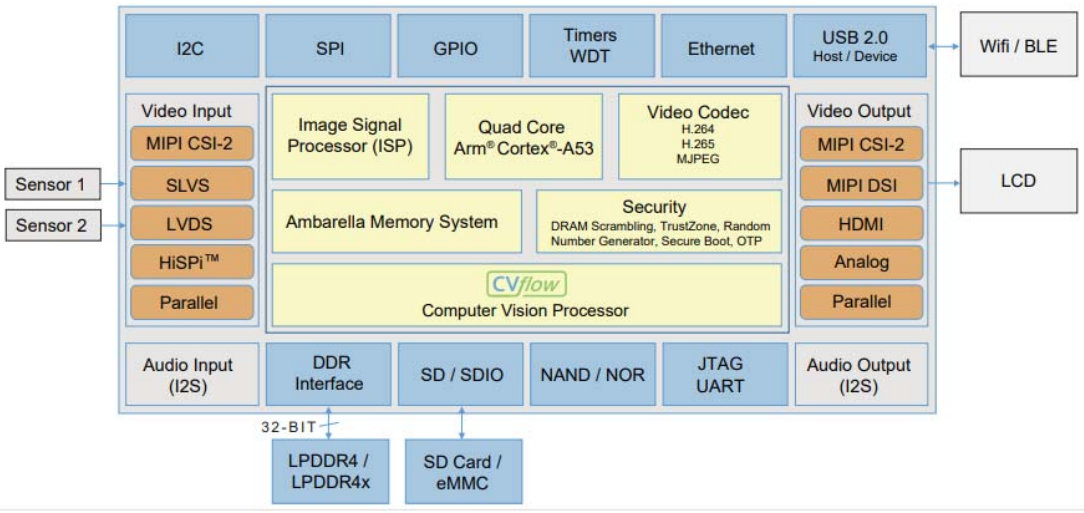
High-Efficiency Video Encoding
- H.265 and H.264 video compression
- Flexible multi-streaming capability
- 8MP30 video performance
- Multiple CBR and VBR bit rate control modes
- Smart H.264 and H.265 encoder algorithms

Target Applications
- Multi-channel drive recorder
- AVM / surround view with auto parking assist (APA)
- Single- / multi-channel eMirror
- ADAS
- Driver / cabin monitoring system (DMS / CMS)

Example : Ambarella ISP+CV: CV22AQ

Block Diagram

The diagram below illustrates a design based on the Ambarella CV22AQ device.



참고 : GEO SEmiconductor
https://www.geosemi.com/applications/

Report 4: 다음 용어 설명 요약

- 다음 용어를 인터넷에서 찾아 이해한 다음 요약하여 정리하시오.
  - MIPI
  - AEC-Q100 Grade 1/2/
  - ISO26262

- 제출 방법
  - 레포트 작성은 손으로 작성하여 아래 한글에 붙여 넣기하여 파일로 제출 하세요. 분량은 2페이지 이내로 하시오.
  - 레포트 제출은 강의지원에 제출
  - 파일명은 report4_성명.hwp

Thank You
Next : LiDAR Concept

